

Лабораторная работа №9

Управление SELinux

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Цель работы

Получить навыки работы с контекстом безопасности и политиками SELinux.

Ход выполнения работы

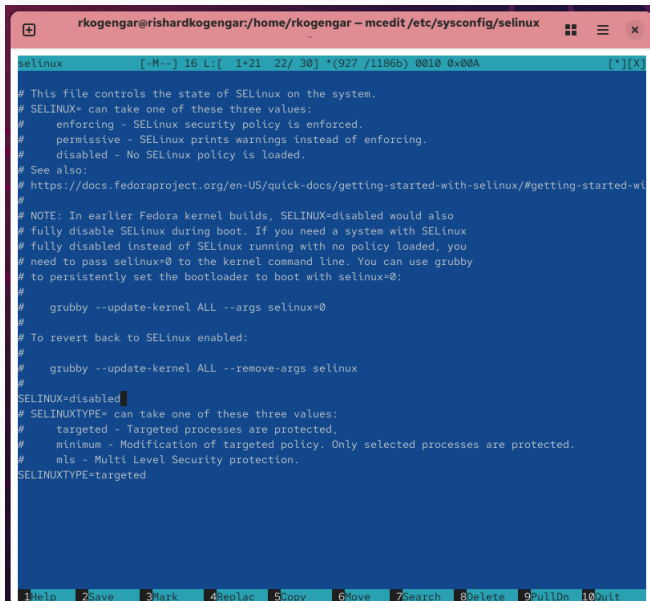
Проверка статуса SELinux

```
rkogengar@rishardkogengar:/home/rkogengar
+
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar# sestatus -v
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:        /etc/selinux
Loaded policy name:              targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33

Process contexts:
Current context:                unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
Init context:                   system_u:system_r:init_t:s0
/usr/sbin/sshd                  system_u:system_r:sshd_t:s0-s0:c0.c1023

File contexts:
Controlling terminal:          unconfined_u:object_r:user_devpts_t:s0
/etc/passwd                    system_u:object_r:passwd_file_t:s0
/etc/shadow                    system_u:object_r:shadow_t:s0
/bin/bash                      system_u:object_r:shell_exec_t:s0
/bin/login                     system_u:object_r:login_exec_t:s0
/bin/sh                        system_u:object_r:bin_t:s0 -> system_u:object_r:shell_exec_t:s0
/sbin/agetty                   system_u:object_r:getty_exec_t:s0
/sbin/init                    system_u:object_r:bin_t:s0 -> system_u:object_r:init_exec_t:s0
/usr/sbin/sshd                 system_u:object_r:sshd_exec_t:s0
root@rishardkogengar:/home/rkogengar# getenforce
Enforcing
root@rishardkogengar:/home/rkogengar# setenforce 0
root@rishardkogengar:/home/rkogengar# getenforce
Permissive
root@rishardkogengar:/home/rkogengar#
```

Проверка статуса SELinux



The screenshot shows a terminal window with the mcedit editor open to the file `/etc/sysconfig/selinux`. The window title is `rkogengar@rishardkogengar:/home/rkogengar - mcedit /etc/sysconfig/selinux`. The terminal output shows the following content:

```
selinux [~M~] 16 L:[ 1+21 22/ 30] *(927 /1186b) 0010 0x00A [*][X]

# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
# See also:
# https://docs.fedoraproject.org/en-US/quick-docs/getting-started-with-selinux/#getting-started-wi
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#   grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#   grubby --update-kernel ALL --remove-args selinux
#
SELINUX=disabled
# SELINUXTYPE= can take one of these three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are protected.
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

At the bottom of the terminal, there is a status bar with the following text: `1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit`.

```
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar# getenforce
Disabled
root@rishardkogengar:/home/rkogengar# setenforce 1
setenforce: SELinux is disabled
root@rishardkogengar:/home/rkogengar# █
```

Рис. 3: Отключение SELinux в конфигурационном файле

Отключение и повторное включение SELinux

```
Booting 'Rocky Linux (6.12.0-55.12.1.el10_0.x86_64) 10.0 (Red Quartz)'  
[ 1.562377] vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on  
an unsupported hypervisor.  
[ 1.562379] vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely b  
roken.  
[ 1.562380] vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported g  
raphics device to avoid problems.  
[ 7.765831] selinux-autorelabel[828]: *** Warning -- SELinux targeted policy relabel is required.  
[ 7.765892] selinux-autorelabel[828]: *** Relabeling could take a very long time, depending on file  
[ 7.765915] selinux-autorelabel[828]: *** system size and speed of hard drives.  
[ 7.768015] selinux-autorelabel[828]: Running: /sbin/fixfiles -T 0 restore
```

Рис. 4: SELinux отключён

Отключение и повторное включение SELinux

```
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar# sestatus -v
SELinux status:                enabled
SELinuxfs mount:               /sys/fs/selinux
SELinux root directory:        /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:         enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Memory protection checking:    actual (secure)
Max kernel policy version:     33

Process contexts:
Current context:                unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
Init context:                   system_u:system_r:init_t:s0
/usr/sbin/sshd                  system_u:system_r:sshd_t:s0-s0:c0.c1023

File contexts:
Controlling terminal:          unconfined_u:object_r:user_devpts_t:s0
/etc/passwd                     system_u:object_r:passwd_file_t:s0
/etc/shadow                     system_u:object_r:shadow_t:s0
/bin/bash                      system_u:object_r:shell_exec_t:s0
/bin/login                     system_u:object_r:login_exec_t:s0
/bin/sh                        system_u:object_r:bin_t:s0 -> system_u:object_r:shell_exec_t:s0
/sbin/agetty                   system_u:object_r:getty_exec_t:s0
/sbin/init                     system_u:object_r:bin_t:s0 -> system_u:object_r:init_exec_t:s0
/usr/sbin/sshd                 system_u:object_r:sshd_exec_t:s0
root@rishardkogengar:/home/rkogengar#
```

Восстановление контекста безопасности

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# ls -Z /etc/hosts  
system_u:object_r:net_conf_t:s0 /etc/hosts  
root@rishardkogengar:/home/rkogengar# cp /etc/hosts ~/  
root@rishardkogengar:/home/rkogengar# ls -Z ~/hosts  
unconfined_u:object_r:admin_home_t:s0 /root/hosts  
root@rishardkogengar:/home/rkogengar# mv ~/hosts /etc  
mv: overwrite '/etc/hosts'? y  
root@rishardkogengar:/home/rkogengar# ls -Z ~/hosts  
ls: cannot access '/root/hosts': No such file or directory  
root@rishardkogengar:/home/rkogengar# ls -Z /etc/hosts  
unconfined_u:object_r:admin_home_t:s0 /etc/hosts  
root@rishardkogengar:/home/rkogengar# restorecon -v /etc/hosts  
Relabeled /etc/hosts from unconfined_u:object_r:admin_home_t:s0 to unconfined_u:object_r:net_conf_  
t:s0  
root@rishardkogengar:/home/rkogengar# ls -Z /etc/hosts  
unconfined_u:object_r:net_conf_t:s0 /etc/hosts  
root@rishardkogengar:/home/rkogengar# touch /.autorelabel  
root@rishardkogengar:/home/rkogengar#
```

Рис. 6: Восстановление контекста безопасности и подготовка к перемаркировке

Настройка контекста для веб-сервера

```
Total                                     4.2 MB/s | 1.6 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : lynx-2.9.0-6.el10.x86_64        1/1
  Running scriptlet: lynx-2.9.0-6.el10.x86_64        1/1

Installed:
  lynx-2.9.0-6.el10.x86_64

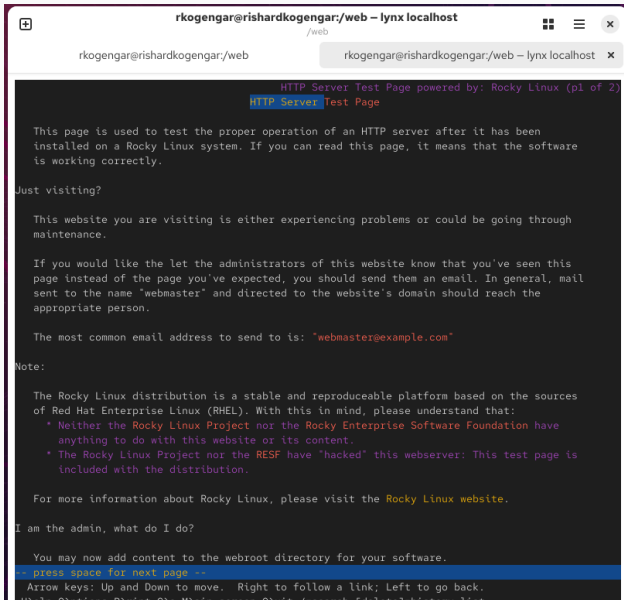
Complete!
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# mkdir /web
root@rishardkogengar:/home/rkogengar# cd /web
root@rishardkogengar:/web# echo "Welcome to my web server" > index.html
root@rishardkogengar:/web#
```

Рис. 7: Создание каталога и файла для веб-сервера

```
#  
# DocumentRoot: The directory out of which you will serve your  
# documents. By default, all requests are taken from this directory, but  
# symbolic links and aliases may be used to point to other locations.  
#  
#DocumentRoot "/var/www/html"  
  
DocumentRoot "/web"  
  
<Directory "/web">  
    AllowOverride None  
    Require all granted  
</Directory>
```

Рис. 8: Изменение DocumentRoot и настроек доступа

Настройка контекста для веб-сервера

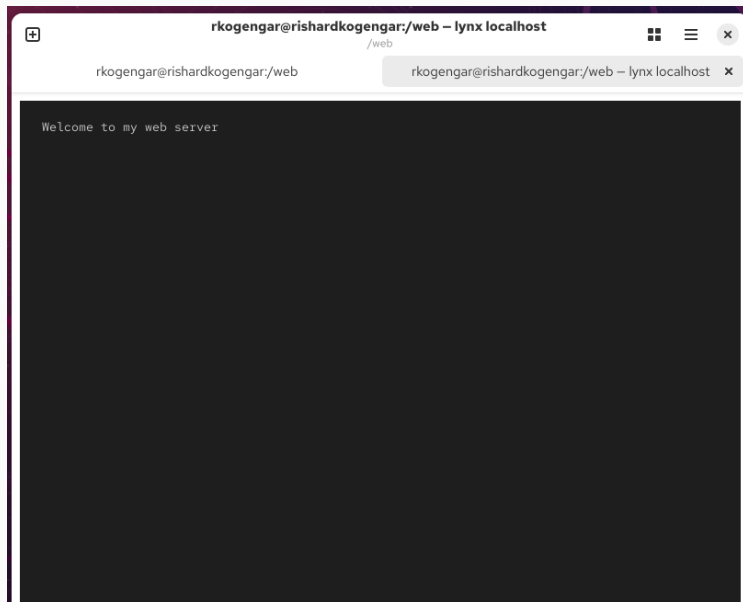


Применение контекста httpd_sys_content_t

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# mkdir /web  
root@rishardkogengar:/home/rkogengar# cd /web  
root@rishardkogengar:/web# echo "Welcome to my web server" > index.html  
root@rishardkogengar:/web#  
root@rishardkogengar:/web# mcedit /etc/httpd/conf/httpd.conf  
  
root@rishardkogengar:/web#  
root@rishardkogengar:/web# systemctl start httpd  
root@rishardkogengar:/web# systemctl enable httpd  
root@rishardkogengar:/web#  
root@rishardkogengar:/web# semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"  
root@rishardkogengar:/web# restorecon -R -v /web  
Relabeled /web from unconfined_u:object_r:default_t:s0 to unconfined_u:object_r:httpd_sys_content_t:s0  
Relabeled /web/index.html from unconfined_u:object_r:default_t:s0 to unconfined_u:object_r:httpd_sys_content_t:s0  
root@rishardkogengar:/web# systemctl restart httpd  
root@rishardkogengar:/web# █
```

Рис. 10: Назначение контекста безопасности и перезапуск службы

Применение контекста httpd_sys_content_t



Работа с переключателями SELinux

```
root@rishardkogengar:/web#  
root@rishardkogengar:/web# getsebool -a | grep ftp  
ftpd_anon_write --> off  
ftpd_connect_all_unreserved --> off  
ftpd_connect_db --> off  
ftpd_full_access --> off  
ftpd_use_cifs --> off  
ftpd_use_fusefs --> off  
ftpd_use_nfs --> off  
ftpd_use_passive_mode --> off  
httpd_can_connect_ftp --> off  
httpd_enable_ftp_server --> off  
tftp_anon_write --> off  
tftp_home_dir --> off  
root@rishardkogengar:/web# semanage boolean -l | grep ftpd_anon  
ftpd_anon_write (off , off) Allow ftpd to anon write  
root@rishardkogengar:/web# setsebool ftpd_anon_write on  
root@rishardkogengar:/web# getsebool ftpd_anon_write  
ftpd_anon_write --> on  
root@rishardkogengar:/web# semanage boolean -l | grep ftpd_anon  
ftpd_anon_write (on , off) Allow ftpd to anon write  
root@rishardkogengar:/web# setsebool -P ftpd_anon_write on  
root@rishardkogengar:/web# semanage boolean -l | grep ftpd_anon  
ftpd_anon_write (on , on) Allow ftpd to anon write  
root@rishardkogengar:/web#
```

Рис. 12: Работа с переключателями SELinux для FTP

Итоги работы

В ходе работы были изучены принципы управления политиками безопасности SELinux, настройка контекстов доступа и использование переключателей для системных служб.